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PERSONALITY AND IMPULSE CONTROL DISORDERS (R LEE, SECTION EDITOR)

Trust Beliefs, Biases, and Behaviors in Borderline Personality Disorder: Empirical Findings and Relevance to Epistemic Trust

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Abstract

Purpose of Review This review summarizes empirical research on trust in BPD, including three primary areas: the prevalence of paranoia, trustworthiness appraisals, and trust-related behaviors in economic exchange paradigms. Connections to the largely theoretical study of epistemic trust in BPD are highlighted.

Recent Findings In trust appraisal paradigms, people with BPD have a bias to rate others as untrustworthy. In behavioral exchange games, they report lower trust in partners and are more likely to rupture cooperation. Recent research has suggested potential explanations for these findings, including differences in affective processing, aberrant social norms and expectations, and difficulty attending to and incorporating social cues.

Summary People with BPD commonly experience paranoia, generally regard others as untrustworthy, and act accordingly. Further research is needed to understand the mechanisms of altered trust processing and to integrate empirical research with recent theoretical research on epistemic trust.

Keywords Borderline personality disorder; trust · Cooperation · Epistemic trust

Introduction

In every interaction—from sharing an elevator with a stranger to discussing relationship difficulties with a spouse—trust enables us to proceed in an inherently uncertain world. To trust is to make oneself interpersonally vulnerable, with the expectation and/or hope that others will act with positive intentions [1]. Trust is central to interpersonal interactions across contexts, and particularly relevant to close relationship functioning [2]. Trust difficulties predict lower relationship satisfaction [3], greater variability in perceptions of romantic partners [4], and lower perceived closeness to partners [5].

Difficulties with trust are particularly salient in people with borderline personality disorder (BPD). Although typical descriptions of BPD do not feature trust impairments, they are implied in the disorder's diagnostic criteria. BPD includes temporary, stress-related paranoia, as well as significant fears of interpersonal abandonment and unstable representations of relationship partners [6]. In the Alternative Model of Personality Disorders [6], BPD is defined in part by high trait antagonism, akin to low agreeableness, a trait domain that includes trust [7].

Trust difficulties in people with BPD are of longstanding clinical interest [8–16]. Research on trust in BPD has focused predominantly on three main areas: (1) paranoia and suspiciousness; (2) appraisals of the trustworthiness of others; and (3) trust-related behaviors in economic exchange paradigms and the influence of oxytocin on these behaviors. A consistent message emerges from these studies: people with BPD commonly consider others untrustworthy and act accordingly. In this review, we consider recent empirical findings related to trust processing in BPD across these three areas of study—beliefs, biases, and behavior. Additionally, we consider an emerging body of largely theoretical research on epistemic trust, and suggest how empirical findings support its relevance to BPD.

Paranoia and Mistrust Beliefs

The term "borderline" was originally coined to connote that the disorder exists in a liminal space between neurosis and

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psychosis [17]. Although many researchers and clinicians currently use the term without meaning to imply this theoretical stance on borderline pathology, the relevance of psychosis to BPD remains. Quasi-psychotic thought has been called "virtually pathognomonic" in BPD because of its 100% prevalence in one study [18]. The experience of quasi-psychotic thought in BPD often centers on nondelusional paranoia (NDP), which includes interpersonal distrust and a belief in the malevolent intentions of others [18–20]. NDP strongly predicts poorer personal and social functioning in BPD [19] and mediates the association of early maltreatment and suicide potential [21]. Although the experience of psychosis and paranoia is often regarded as nondelusional, it warrants mention that a smaller proportion of those with BPD do experience paranoid delusions [19].

In one longitudinal study, Zanarini et al. [20] found that, at baseline, 86.6% of people with BPD endorsed NDP, compared with 43.1% of people with other personality disorders (PDs). Although NDP decreased significantly for both groups over follow-up assessments, the BPD group endorsed NDP at more than twice the rate (43.3%) of those with other PDs (20.7%) 16 years later. More recent work has found that although paranoia is not uncommon in other personality disorders, it is particularly associated with risk for BPD [22]. Moreover, paranoia is more prevalent and severe in BPD compared with dysthymia [23] and even schizophrenia [19].

Given that the experience of paranoia is more common in BPD and purportedly "transient" and "stress-related" [6], one might expect people with BPD to have a greater degree of paranoid reactivity in the context of stress, relative to other groups. However, there is limited and mixed evidence for this difference. In an experience sampling study, Glaser et al. [24] found that people with BPD showed the greatest psychotic reactivity to daily stress compared with people with Cluster C PDs, people with psychotic disorders, and healthy controls. However, although specifically *paranoid* reactivity was greater for patient groups relative to healthy controls, it did not differ between patient groups.

Perhaps unsurprisingly given the prevalence of NDP in BPD, schemas that include mistrust of others figure prominently in the disorder. These include distrust [25], mistrust/abuse [26, 27], and disconnection/rejection [28]. The prevalence of mistrust/abuse beliefs may differentiate BPD from other personality disorders (e.g., [26, 29]), and it has been suggested that the predominance of mistrustfulness/anger beliefs may define a subtype of BPD [30].

The specific manner in which paranoia and mistrust beliefs in BPD manifest in social judgments and interactions is not clear from studies of prevalence and severity alone. To better understand and isolate trust processes in people with BPD, researchers have turned to the lab, where biases in basic trust appraisals consistent with paranoia and mistrust are clear.

Trust Appraisals

In general, when provided with limited information—photographs or short videos—about others, people with BPD describe them more negatively compared with healthy controls and people with Cluster C PDs [49], and as more aggressive compared with healthy controls and people with unipolar depression [50]. For trustworthiness judgments specifically, it is now a well-replicated finding that people with BPD make more negative appraisals than other groups (e.g., [31, 32•, 33, 35, 36•]). Trust appraisal findings are summarized in Table 1.

In an early study on trustworthiness appraisals in BPD, people with BPD rated faces as less trustworthy and less approachable compared with healthy controls [35]. Using similar stimuli, Fertuck et al. [31] showed participants a series of faces that varied on either a fear or trustworthiness dimension, and asked them to rate the fearfulness or trustworthiness of each face. Compared with controls, people with BPD had a slower response time for faces with ambiguous trustworthiness and generally rated male faces as less trustworthy but no more or less fearful. Their responses were consistent with a negative trustworthiness response bias, not with greater sensitivity to cues of trustworthiness or ability to discriminate trustworthy from untrustworthy faces. Negative response bias operated independently of fear appraisals, indicating that trustworthiness bias does not result from difficulty with fear recognition.

Some research has examined how negative emotional context influences trustworthiness appraisals. Although an induced negative emotional state does not seem to alter how people with BPD appraise the general malevolence of others [51], negative emotional primes may unduly influence trustworthiness appraisals for people with BPD. In a non-clinical sample, Masland and Hooley [32•] asked participants to rate the trustworthiness of faces following neutral, negative, or positive affective primes. Regardless of prime condition, a BPD features group (people with three or more BPD criteria) rated faces as less trustworthy relative to a control group of people with two or fewer BPD criteria. This finding is consistent with a negative response bias. The BPD features group was also more greatly influenced by negative primes than the control group. Negative affect may augment the trust appraisal bias observed in BPD, and negative affective processes may shed light on affective mechanisms through which BPD influences trust.

Another potential mechanism is rejection sensitivity: a readiness to expect, perceive, and overreact to signs of rejection [52]. People with BPD have significant rejection sensitivity on self-report (e.g., [53, 54]) and behavioral (e.g., [55]) measures. In non-clinical samples, rejection sensitivity mediates the association of BPD features with negative trustworthiness appraisals [33, 36•]. More specifically, the emotional components of rejection sensitivity (i.e., anxiety and anger related to rejection) mediate the association, while the cognitive

Table 1 Summary of trust appraisal and behavioral exchange studies

Study type	Authors	Relevant task description	Primary/selected findings	General limitations
Trust appraisal	Fertuck et al. [31]	Rate faces on dimensions of trustworthiness and fear	BPD trust appraisals < HC but only for male faces; accounted for by response bias (not increased sensitivity or discriminability); trust appraisal independent of fear appraisal	72% female; 56% white; face stimuli all male; significant comorbidity; group medication differences
	Masland and Hooley [32•]	Rate faces on trustworthiness; each face preceded by neutral, negative, or positive prime	Negative trust appraisal bias in BPD; unduly influenced by negative primes	73% female; 65% white; non-clinical sample; significant comorbidity; medications unreported; stimuli race unreported
	Miano et al. [33]	Rate faces on 17 dimensions, including trustworthiness	Rejection sensitivity mediates association of BPD features with lower trust appraisals	College students; 69% female; comorbidity and medications unreported; face stimuli 100% white
	Miano et al. [34•]	Rate romantic partner trustworthiness following discussions of neutral topics (favorite films) and threatening topics (relationship threat or personal fears)	BPD trust appraisals = HC after neutral topic; BPD appraisals after threatening topics < neutral topic; HC appraisals after only relationship threat < neutral topic	Heterosexual couples only; married couples only; race/ethnicity unreported; significant comorbidity; medications unreported
	Nicol et al. [35]	Rate faces on 6 dimensions, including trustworthiness	BPD trust and approachability appraisals < HC; approachability but not trustworthiness associated with childhood trauma in BPD group	76% female; race/ethnicity unreported; significant comorbidity; group medication differences; face stimuli 100% white
	Richetin et al. [36•]	Rate faces on trustworthiness	Emotional, but not cognitive components of rejection sensitivity mediate association of BPD features with lower trust appraisals	College students; 100% female; comorbidity and medications unreported; face stimuli 100% white
Behavioral exchange	Botsford et al. [10]	Development of interpersonal trust scale; traditional trust game	BPD monetary units transferred < HC; BPD scored < HCs, MDD, SAD on interpersonal trust measure; interpersonal trust associated with therapeutic alliance only for BPD	73% female in trust game analysis; race/ethnicity unreported; comorbidity unreported; medications not assessed
	Fineberg et al. [37•]	Social valuation task: requires learning about reward probability of nonsocial and social (from an actor) cues	BPD weighted social cues more heavily; BPD social learning rates slower than HC; BPD less sensitive to social volatility	100% female; 49% white; comorbidity unreported; group medication differences; task actors 100% white/female
	Franzen et al. [38]	Traditional trust game with four trustees; varied fairness and emotional cues associated with repayment ratio	BPD but not HC used fairness to guide behavior in presence of emotional cues; BPD = HC on emotion recognition; BPD = HC on trustee fairness ratings	73% female; race/ethnicity unreported; comorbidity unreported; group medication differences; trustees 100% white
	Hepp et al. [39]	Raters evaluated likeability and trustworthiness of people with BPD and HC; dictator game	People with BPD rated as less likeable and trustworthy than HC; BPD = HC on cooperation in dictator game	Significant comorbidity; race/ethnicity unreported; group medication differences; BPD group socioeconomic status < HC

Table 1 (continued)

Study type	Authors	Relevant task description	Primary/selected findings	General limitations
	Hula et al. [40•]	Reanalysis of traditional trust game data from King-Casas et al. [41•]; computational modeling of risk aversion and irritation	Identified subgroup of "perilous" trustees predominantly from BPD group who had difficulty recognizing/adapting to investor irritation; investor game captures trust not general risk aversion	See King-Casas et al. [41•]
	King-Casas et al. [41•]	Traditional trust game	Relative to HC, BPD reported less trust in investors; made smaller repayments; BPD did not "coax" to repair cooperation ruptures; anterior insula differentiates BPD from HC during game	Groups matched on sex, age, education, IQ but demographics not reported; comorbidity and medications unreported
	Liebke et al. [42•]	Randomly assigned to experience rejection or acceptance in simulated interaction before playing modified trust game	BPD expectations of acceptance < HC; BPD adjusted to negative but not positive feedback; BPD less cooperative in trust game after acceptance	100% female; race/ethnicity unreported; significant comorbidity
	Preuss et al. [43]	Trust game with social and non-social risk; punishment/dictator game; cooperation game	BPD investment inconsistency > HC and MDD for social risk conditions of trust and punishment games; inconsistency associated with perceived social support; all groups increased investment over course of trust and punishment games	100% female; race/ethnicity not reported; significant comorbidity; group medication differences; MDD age > BPD
	Thielmann et al. [44]	Dictator game; ultimatum game	BPD features predicted reactive cooperation breaches in ultimatum game, mediated by agreeableness; BPD features not associated with active breaches in dictator game	Online sample; 82% female; race/ethnicity not reported; comorbidity and medication not reported
	Unoka et al. [45]	Traditional trust game; risk/lottery game	BPD monetary transfers < HC and MDD in trust game but = HC and MDD in risk/lottery game; MDD = HC in both games; BPD negative predictions of outcomes > HC and MDD in trust but not risk game; MDD negative predictions of outcomes > HC for risk but not trust game	80% female; race/ethnicity not reported; significant comorbidity in MDD group (unreported in BPD group)
Behavioral exchange with pharmacological intervention	Bartz et al. [46•]	Randomly assigned to intranasal OXT or placebo before assurance game (variation of Prisoner's Dilemma)	BPD expected less cooperation from partners; BPD more likely to defect in OXT vs. placebo; HC demonstrated opposite pattern; attachment anxiety moderated effect of OXT (OXT had greatest effect on trust and cooperation for anxiously attached)	60% female

Table 1 (continued)

Study type	Authors	Relevant task description	Primary/selected findings	General limitations
	Ebert et al. [47•]	Randomly assigned to intranasal OXT or placebo before modified trust game with attractive and unattractive images of virtual trustees	BPD transferred less in OXT vs. placebo; BPD transferred more money to attractive vs. unattractive trustees in OXT and placebo (HC showed attractiveness bias in OXT only); childhood neglect associated with transfer size for BPD only after OXT	70% female; hormonal contraceptives required; no control for menstrual phase; medications uncontrolled
	Roberts et al. [48]	Randomly assigned to liquid ACT or placebo before traditional trust game	BPD features associated with increased trust in ACT vs. placebo; those with low BPD features showed no effect of ACT vs. placebo	College students; race/ethnicity not reported; comorbidity and medication not reported

HC, healthy controls; *MDD*, major depressive disorder; *SAD*, social anxiety disorder; *OXT*, oxytocin; *ACT*, acetaminophen

components (i.e., expectations of rejection) do not [36•]. This again suggests that emotional processes contribute to trustworthiness appraisal.

The assessment of basic trustworthiness appraisals is conducive to experimentation, yet its ecological validity is questionable. Notably, trustworthiness appraisals pair participants with strangers, failing to account for the effect of relationship intimacy. In a rare study matching participants with close others, Miano et al. [34•] examined trust appraisals in two sets of unmarried heterosexual couples: in one set the women were diagnosed with BPD; in the other set neither partner had BPD. Each couple discussed personal fears, possible partner separation, and favorite films. Trustworthiness appraisals of partners made after the discussion of favorite films did not differ for women with BPD versus healthy controls. Women with BPD showed a significant reduction in trustworthiness (but not confidence) appraisals of their partners following the threatening discussions. This is further evidence that general negative emotional context negatively impacts how women with BPD consider trustworthiness.

Behavioral Exchange Paradigms

The "trust game" is the most widely used and adapted behavioral paradigm for examining trust and cooperation in BPD. Borrowed from behavioral economics, it tracks financial decision-making in a simulated interaction [41•]. Numerous trust game studies have reported decreased trust and cooperation in people with BPD [10, 41•, 45]. Findings from behavioral exchange paradigms are summarized in Table 1.

King-Casas et al. [41•] were the first to study trust game behavior in people with BPD. They employed a multi-round economic exchange game in which an investor and a trustee each begin with a set sum of money. A single round proceeds as follows: the investor decides how much money to send to the trustee, this amount is tripled and transferred to the trustee, the trustee elects what proportion of this updated sum (the initial sum plus the received investment) to return to the investor, and the chosen amount is transferred back to the investor. Cooperation from both parties is mutually beneficial. Investors demonstrate trust in their partner by sending large investments; trustees reciprocate trust by making a large repayment. Compared with healthy trustees, trustees with BPD were less trusting and cooperative in the exchange game [41•]. They reported lower levels of trust in investors and, accordingly, made significantly smaller repayments. Although investors initially transferred the same amount of money to healthy trustees and trustees with BPD—reflecting equal trust in both groups—they transferred significantly less to trustees with BPD in later rounds—as the game progressed, investors lost trust in trustees with BPD due to their uncooperative behavior.

Later trust game studies distinguished active cooperation (fairness, non-exploitation) from reactive cooperation (forgiveness, non-retaliation). Thielmann et al. [44] found that individuals with higher levels of borderline features displayed a greater tendency to retaliate, suggesting that BPD impairs only *reactive* cooperation—hindering the ability to tolerate and forgive. Hepp et al. [39] found no differences in active cooperation between people with BPD and healthy controls. Thus, individuals with BPD do not transfer less money because they are more exploitative or unfair.

Other work has considered that heightened risk aversion, as opposed to diminished interpersonal trust, may explain decreased spending in people with BPD. Unoka et al. [45] conducted both a traditional trust game and a "risk game," in which participants believed that a computer lottery program (instead of a human partner) randomly determined repayments. In the trust game, investors with BPD transferred significantly less money to trustees and predicted worse outcomes. However, in the risk game, investors with BPD and controls transferred comparable amounts of money. Together, these findings demonstrate that decreased spending specifically reflects diminished interpersonal trust. The finding that BPD affects trusting behavior and not risk aversion has been replicated [40•, 43].

Trust game data establish that people with BPD are less trusting and suggest explanations for why this is the case. One possible explanation, supported by multiple studies, is the possession of atypical social norms [41•, 42•, 45]. While playing an economic exchange game, people with BPD exhibited differential activity in the bilateral anterior insula, which is known to respond to social norm violations [41•]. In healthy controls, bilateral anterior insula activity increased after receiving small investments, which violated the social norms of trust and cooperation. Large, norm-adhering investments provoked less insula activity. Unlike healthy trustees, those with BPD showed similar insula activity in response to both low and high investments. As they hold distorted, overly negative social expectations, trustees with BPD may anticipate partner betrayal and, therefore, may perceive low investments as normative.

Critically, evidence from trust games suggests that people with BPD predict partner betrayal regardless of actual partner behavior. Liebke et al. [42•] found that following rejection, both healthy controls and people with BPD adjusted their expectations to predict future rejection. However, following acceptance, only the healthy controls adjusted their expectations to predict more positive feedback. People with BPD did not become more trusting after an experience of social acceptance. In fact, people with BPD who had been socially accepted were actually less trusting of cooperative trustees. Liebke et al. [42•] speculated that the ability to learn from positive social experiences—and to adjust one's social expectations and behaviors accordingly—may be compromised in people with BPD.

The inability to use partner feedback to adjust behavior is disadvantageous when a partner is trusting: the failure to reciprocate ruptures trust in the partnership. This inability is also disadvantageous when a partner is distrustful, as it prevents trust from being repaired. In economic exchange games, participants have the opportunity to repair trust by "coaxing." When they receive a low investment—a cue that they have ruptured trust with their investor—participants can choose to repay a high proportion to demonstrate their renewed trustworthiness and secure larger investments in subsequent rounds. Healthy trustees are more likely than trustees with BPD to coax following a low investment [41•]. Trustees with BPD do not change their behaviors in order to regain investor trust and repair the alliance. In a re-analysis of the data set from King-Casas et al. [41•], Hula et al. [40•] identified difficulties adapting to investor irritation in a subgroup of people from both BPD and healthy control groups, whom they labeled "perilous." A significantly larger proportion of these perilous trustees came from the BPD group. Perilous trustees were more likely to rupture trust with their investors and to fail to repair the alliance.

Traditional trust games are not the only context in which people with BPD experience difficulties with social learning. Using the Social Valuation Task (SVT), an economic decision-making game in which participants receive trustworthy and untrustworthy advice from an actor posing as another participant, Fineberg et al. [37•] found that although people with BPD were more attentive to social (and nonsocial cues), they had slower social learning rates compared with healthy controls. In conditions where actor advice was particularly volatile, people with BPD adapted their learning more slowly and were less sensitive to increased volatility. Fineberg et al. [37•] suggested that people with BPD may expect greater volatility (or interpersonal unreliability), and thus are not as surprised by or as sensitive to rapid changes in behavior or trust-related cues from others. In further support of this idea, people with BPD exhibited less implicit distress upon learning that their partner was actually an actor in the post task debriefing. Confirmation of partner deception seemingly reduced distress in people with BPD.

People with BPD may be less able to adapt when they encounter volatility and irritability because these cues confirm their expectations. Alternatively, impairments in mentalizing, the ability to understand the mental states of oneself and others [56], may explain apparent difficulties incorporating social cues in trust-related behaviors. Although some argue that trust game data indicates superior mentalizing in people with BPD [38], BPD is commonly understood as a disorder of impaired mentalizing [56]. On a neural level, disrupted insula signaling in people with BPD could hinder awareness that a relationship is in jeopardy and in need of repair [57].

Variance in levels of awareness and cooperation among people with BPD demonstrates that trusting behaviors across individuals with BPD are not uniform, and highlights the importance of individual differences. Although Hula et al. [40•] found that "perilous" trustees were overrepresented in the BPD group, it is crucial to note that not all individuals with BPD qualified as "perilous." In fact, with the exception of this subgroup, no significant behavioral differences were found between people with BPD and healthy controls. Hula et al. [40•] proposed the development of a "perilous" clinical phenotype to distinguish this subgroup categorically from people with BPD who are more cooperative. From a dimensional standpoint, personality traits associated with BPD (i.e., high neuroticism and low agreeableness) also predict a failure to repair broken trust in exchange games [44, 58]. Individual differences with respect to personality traits should be further considered.

The Effect of Oxytocin

Because the neuropeptide oxytocin (OXT) increases trusting behavior in healthy individuals [59–61], it seems to hold some promise for treating disorders in which interpersonal trust and cooperation are compromised, including BPD. However, across trust game studies, intranasal OXT administration actually reduces trust in people with BPD [46•, 47•]. In a game requiring mutual cooperation (as opposed to defection) for maximum monetary outcome, people with BPD who received OXT trusted their partners significantly less than people with BPD who received a placebo [46•]. Inpatients with BPD were also less trusting in a traditional economic exchange game when administered OXT [47•].

OXT differentially affects healthy controls and people with BPD. However, even among people with BPD, the effect of OXT is moderated by individual differences, including variation in attachment anxiety and attachment avoidance. In one study, "fearful" individuals with both high attachment anxiety and avoidance were less cooperative when administered OXT, while "preoccupied" individuals with high anxious attachment and low attachment avoidance were more cooperative when administered OXT [46•]. Additionally, childhood adversity may moderate OXT's effects. In a trust game, inpatients with a history of early emotional neglect transferred significantly less money when administered oxytocin compared with placebo [47•].

There are several possible explanations for the divergent effects of OXT on trust, both among people with BPD and between people with BPD and healthy controls. First, OXT may increase the salience of social cues (both positive and negative) rather than solely positive social emotions (e.g., trust; [56, 59•). Therefore, for people with BPD, anxious attachment, and/or a history of childhood trauma, exogenous OXT may exacerbate negative social emotions, which, in turn, lower trust. Second, if OXT increases feelings of connectedness, it may inspire fears of abandonment and remembrances of past interpersonal conflict for these same populations [46•]. OXT increases both in-group connection and out-group aggression [59]. Given difficulties with splitting and identity instability in BPD, it is possible that the distinction between in-group and out-group may be blurred in moments of distress. Accordingly, OXT administration may increase aggression towards close others during interpersonal conflict. While this idea has yet to be investigated, it is consistent with the finding that, whereas healthy controls report greater trust in closer partners, people with BPD make no distinctions between peripheral and central members of their social networks in trust appraisals [62]. Lastly, people with BPD may have a dysregulated OXT system that reacts differently to exogenous OXT [46•]. Most likely, some combination of these factors—in addition to others—underlies differential responses to OXT administration for people with BPD.

Recently, research has expanded beyond OXT to examine the effect of the drug acetaminophen on individuals with BPD. In people with high BPD features, acetaminophen reduces distrusting behavior in economic exchange games [48]. Future research should investigate treatment interventions—both pharmacological and psychotherapeutic—with the potential to promote trust in people with BPD.

Epistemic Trust

We are struck by the considerable unexplored connections between these empirical findings and recent, largely theoretical, work on epistemic trust. Epistemic trust is the belief that knowledge gained through social learning is authentic, relevant to oneself, and generalizable [63•]. The development of epistemic trust enables the transmission of important social and cultural information. Difficulties with epistemic trust interfere with salutogenesis, the ability to derive positive benefit from social experiences [8, 64]. Lack of salutogenesis creates generalized issues with resilience to life stressors, which predispose an individual to psychopathology broadly and to BPD in particular [8, 65].

The common experience of paranoia and the bias to rate others as less trustworthy are congruent with the general manifestation of epistemic mistrust, which includes the assumption of malevolent intentions of others [63•]. Behavioral exchange studies are also consistent with epistemic mistrust and lack of salutogenesis in BPD. For example, the failure to update betrayal expectations after the experience of acceptance [42•] demonstrates a failure to update existing belief systems in a manner that would be beneficial, as does the failure to invest in others or heed their advice, even when doing so would result in the best outcome.

Moreover, developmental research and theory suggests that epistemic trust requires openness to ostensive cues from interaction partners [66]. To experience salutogenesis, one must be receptive to the social cues that indicate an interaction partner is attempting to share information. Empirical findings on both trustworthiness appraisal and behavior in exchange games suggest impaired ability to use social cues for personal benefit. In multiple studies, people with BPD were seemingly insensitive to cues of irritability from others [37•, 40•, 41•]. They may also use emotional cues to a lesser extent [38], or use inappropriate cues, including partner attractiveness [47•]. It is possible that these findings are related to epistemic vigilance—the essential belief or expectation that others have misinformation [67]. These findings may also help explain why oxytocin, which increases the salience of social cues and should seemingly increase openness to social information, further entrenches mistrust for people with BPD. If social cues are threatening or reminiscent of early learning experiences that resulted in harm, they may signal that epistemic trust is maladaptive.

Evidence that adverse childhood experiences influence trust processing for people with BPD is also consistent with epistemic trust theory. In both trust appraisal studies and behavioral exchange games, childhood adversity predicts trust processing alterations [31, 47•]. In qualitative studies, people with BPD frequently link attachment disruption or abuse to their difficulties trusting others [11, 13]. Attachment disruptions, including childhood neglect or abuse, are theorized to cause epistemic freezing or petrification, in which an individual is hypervigilant to interpersonal hazards and their capacity for trust is shut down [68] in a manner that may be adaptive when the social environment is dangerous [63•, 69].

Recently, MBT thought leaders have argued that the therapeutic relationship in MBT provides a new attachment experience in which patients can learn to build epistemic trust and openness, which may generalize to salutogenesis in social experiences outside of therapy [8, 63•]. Although the empirical study of epistemic trust is limited by the lack of measurement tools [15], initial evidence suggests that enhancing epistemic trust may improve outcomes for people with BPD [70]. Better operationalizations of epistemic trust in adolescents and adults will allow for fruitful integration of the types of empirical research reviewed here and treatment research.

Conclusion

Recent empirical research richly illustrates what has long been known clinically: people with BPD have difficulties trusting and cooperating with others. They commonly struggle with paranoia and/or mistrust beliefs, judge others as untrustworthy, and act accordingly—to their own detriment. However, the interplay of paranoia, beliefs, trust appraisals, trusting behaviors, and epistemic trust remains understudied. So long as these areas of research operate independently of one another, a comprehensive account of trust processing difficulties will remain obscured.

While this complex interplay may be best isolated and controlled in the laboratory, it is essential that trust research becomes more ecologically valid. Although trust games allow players to participate in simulated interactions, these interactions are far from realistic social encounters. Players interact solely with strangers, and the primary consequence of betrayed trust is economic, not social or personal. We urge the incorporation of real relationship partners into laboratory research. Outside of the lab, experience-sampling and daily-diary methods are well-suited to capture how real-world contexts influence trust processing. Research on trust in BPD also suffers from many of the limitations that pervade the broader BPD literature, including gender disparity, significant comorbidity, and medication differences between groups. These are detailed in Table 1.

Methodological limitations and variance notwithstanding, recent empirical efforts have done much to characterize how trust processing differences manifest in people with BPD. Future research should further refine that characterization, and consider the causes—both proximal and distal—and mechanisms underlying differences in trust processing. Additionally, the integration of work on trust beliefs, biases, and behavior is essential to improve not only our basic understanding of trust processing differences, but also patient care and outcomes. One particularly promising avenue for advancing patient care is the integration of these empirical efforts with largely theoretical work on epistemic trust, which may aid the development of trust-enhancing treatments for people with BPD.

Compliance with Ethical Standards

All reported studies/experiments with human or animal subjects performed by the authors have been previously published and complied with all applicable ethical standards (including the Helsinki declaration and its amendments, institutional/national research committee standards, and international/national/institutional guidelines).

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